[scale=0.85]

[font=, black!80] at (10.5,13.2) **Why the Optimiser Chose $n = 12$ Despite Beam-Only Scaling; [font=,**

[shift=(0,0)] [red!3, rounded corners=6pt] (-0.5,-2.0) rectangle (8.0,11.8); [font=, red!60!black] at (3.

//ıin 3/1.00/1.00/0, 6/1.52/1.52/1, 8/1.77/1.77/2, 10/1.99/1.99/3, 12/2.19/2.19/4 * 3.0 1.0 + ı* 1.

[-¿, thick] (0.5,1.5) – (7.8,1.5) node[right, font=] n; /ıin 3/0, 6/1, 8/2, 10/3, 12/4 [font=] at (1.0+ı*

[font=, red!70!black, align=center, fill=white, draw=red!30, rounded corners=3pt, inner sep=4pt] a
n=3 is lightest; [-¿, red!50, line width=1.5pt] (4.5,4.5) – (1.5,3.0);

[-¿, line width=4pt, ¿=stealth, magenta!60] (8.4,6.5) – (9.8,6.5); [font=, magenta!60, align=center] a
system
effects;

[shift=(10.5,0)] [blue!2, rounded corners=6pt] (-0.5,-2.0) rectangle (11.0,11.8); [font=, blue!60!black]

[-¿, green!40!black, line width=3pt] (1.5,9.0) – (6.5,4.0); [font=, green!40!black, align=center] at (4.0
discovers
n=12;

[font=, blue!70!black, align=left, fill=white, draw=blue!30, rounded corners=4pt, inner sep=5pt] at
drops $\propto 1/n$. Beams can be thinner.
2. Knuckle coupling: Thinner beams need smaller
knuckles. Knuckle mass scales $\propto D_o \cdot t$.
3. Tether economy: More lines = more tether mass,
but this is green!50!blackoffset by beam+knuckle savings.;

[font=, green!50!black, fill=green!5, draw=green!50!black, rounded corners=5pt, inner sep=6pt] at (

[font=, black!50, align=center] at (5.25,0.8) V6.2 corrected-physics campaign
9 rings, $n_{\mathbf{exp}}=1$, $\beta=-0.13$, 45° bank;

[black!3, rounded corners=6pt] (-0.5,-2.3) rectangle (21.5,-1.2); [font=, black!70, align=center] at (10